

Dexter Fernandes

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Core Technical Stack

- **Perception and Robotics:** PyTorch, TensorFlow, YOLOv5/v8, Vision Transformer (ViT), OpenCV, OpenVINO, ROS2 (Nav2), LiDAR SLAM (Cartographer), Sci-kit Learn
- **Backend & Cloud / Edge:** Python 3, C++, FastAPI, Azure IoT Hub, Intel VPU, NVIDIA Jetson AGX, PostgreSQL
- **DevOps & MLOps:** Git, Docker, Linux, MLFlow, Optuna, DVC, GitHub Actions, Kubernetes

Work Experience

Esbaar, *AI Engineer*

Jan 2023 - Aug 2023

- Developed time-series models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation achieving 23% higher F1-score than human experts.
- Built an automated sensor-data cleansing pipeline (3.6 M data points) that cut experiment-to-deployment turnaround from 3 days to less than 24 hours.
- Led strategic MLOps initiatives to continuously enhance training and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Spearheaded the end-to-end development and production rollout of YOLOv5-based computer-vision models, elevating on-rig worker and equipment detection accuracy by 15% and strengthening site-safety monitoring.
- Refactored a monolith to a Kafka-driven dockerised micro-service mesh failure-isolation MTBF by 53%.
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.
- Participated in formal peer code reviews and quality-assurance activities, ensuring adherence to coding standards and robust, reliable deliverables.

LivNSense Technologies Pvt. Ltd, *Lead Software Engineer - AI*

May 2020 - Dec 2022

- Led a 4-engineer team to deliver ViCAS, a vision-based collision avoidance system, by curating domain-specific datasets, maintaining models in production, and resolving edge-case failures, reducing false positives by 27%.
- Deployed quantised INT8 OpenVINO vision models over Microsoft Azure IoT to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Accelerated vision model inference with mixed-precision quantisation and problem-reframing, reducing latency from 4.5 FPS to 15 FPS, enabling real-time inference and reporting of OSHA violations.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Engineered synthetic training data pipelines using OpenCV and NumPy, reducing edge-case model development time from 6 days to 2 days for high-risk production environments.

Education

Master's Degree in Robotics, *2:1 degree*

Sep 2023 - Sep 2024

University of Bristol

- Engineered a UGV system (Clearpath Husky) with multi-sensor fusion (Velodyne LiDAR, Intel RealSense RGBD camera, Symetrica Verifinder).
- Deployed real-time LiDAR SLAM (ROS2 Nav2 + Velodyne) on Jetson AGX Xavier, containerized with Docker for field-deployable, point-cloud mapping.
- Achieved precise radionuclide localisation using advanced odometry techniques like Scan-to-Map ICP, by significantly reducing drift in real-time LiDAR SLAM operations.
- Implemented a mono camera calibration and point-cloud 3-D reconstruction pipeline in C++/OpenCV for the MSc Robotics thesis, delivering metric-accurate environment mapping for downstream perception tasks.

Bachelor's Degree in Electrical and Electronics

Jul 2016 - May 2020

Visvesvaraya National Institute of Technology